

Concept Question 12.1 Formation of the Solar System

Imagine that the Solar System consists only of gas giant planets. What could you say about the original protoplanetary disk? What would be the difference with the “real” Solar protoplanetary disk?

Problem 12.2 Pb-Pb dating of the early Solar System

Lead (Pb) has four naturally occurring isotopes (^{204}Pb , ^{206}Pb , ^{207}Pb and ^{208}Pb) that were inherited from the molecular cloud in relative abundances of 1 : 9.307 : 10.294 : 29.476. While ^{204}Pb still preserves its primordial abundances, ^{206}Pb , ^{207}Pb and ^{208}Pb have increased in abundance through the radioactive decay of ^{238}U , ^{235}U and ^{232}Th , respectively. Given their half-lives (see table below), the dual decay $^{238}\text{U} \rightarrow ^{206}\text{Pb}$ and $^{235}\text{U} \rightarrow ^{207}\text{Pb}$ can be used to determine the ages of meteorites formed in the first 5 Myr of the Solar System.

Isotope	Half lives [yr]	Decay constant [yr^{-1}]
^{232}Th	1.4010×10^{10}	4.9475×10^{-11}
^{235}U	7.0381×10^8	9.8485×10^{-10}
^{238}U	4.4683×10^9	1.5512×10^{-10}

- a) Write the decay equations for $^{238}\text{U} \rightarrow ^{206}\text{Pb}$ and $^{235}\text{U} \rightarrow ^{207}\text{Pb}$.
- b) Using the decay equations, calculate the current isotopic ratios of $(^{207}\text{Pb}/^{204}\text{Pb})_{\text{today}}$ and $(^{206}\text{Pb}/^{204}\text{Pb})_{\text{today}}$ and combine these ratios to obtain a linear relation of the form:

$$\left(\frac{^{207}\text{Pb}}{^{204}\text{Pb}}\right)_r = a \times \left(\frac{^{206}\text{Pb}}{^{204}\text{Pb}}\right)_r \quad (1)$$

where $(^{20x}\text{Pb}/^{204}\text{Pb})_r$ is the radiogenic ratio:

$$\left(\frac{^{20x}\text{Pb}}{^{204}\text{Pb}}\right)_r = \left(\frac{^{20x}\text{Pb}}{^{204}\text{Pb}}\right)_{\text{today}} - \left(\frac{^{20x}\text{Pb}}{^{204}\text{Pb}}\right)_{\text{original}} \quad (2)$$

- c) The file `Pb_Connelly_2012.dat` contains the $(^{207}\text{Pb}/^{204}\text{Pb})_r$ and $(^{206}\text{Pb}/^{204}\text{Pb})_r$ measurements for the meteorite NWA 5697 (from Connelly et al. 2012). Plot $(^{207}\text{Pb}/^{204}\text{Pb})_r$ vs. $(^{206}\text{Pb}/^{204}\text{Pb})_r$, measure the slope and use this value to find the age t .

Hint: Assume $^{238}\text{U}/^{235}\text{U} = 137.786 \pm 0.011$.

Problem 12.3 Radial velocity measurements

In the lecture, we established the equation

$$k = 28.4 \left(\frac{\text{m}}{\text{s}} \right) \cdot \left(\frac{M_2}{M_J} \right) \sin(i) \cdot \left(\frac{P}{\text{yr}} \right)^{-1/3} \cdot \left(\frac{M_1}{M_\odot} \right)^{-2/3} \cdot \frac{1}{\sqrt{1-e^2}} \quad (3)$$

for the amplitude k of an RV signal.

- a) Express the mass, period, and eccentricity of Earth, Neptune, and Jupiter in units of M_J and yr.
- b) Assume $\sin i = 1$ and calculate the RV effects produced by Earth, Neptune, and Jupiter on the Sun.
- c) We want to develop a new RV instrument to search for exoplanets on perfectly circular orbits - CORVI (Circular Orbit Radial Velocity Instrument, financed by a Swiss-Italian consortium).
 - c1) How precise does CORVI have to be to detect a signal from a Hot Jupiter with $P = 5$ d orbiting a sun-like star at inclinations of up to 45° .
 - c2) What precision do we need if we want to detect Neptune-mass ($M = 17 M_\oplus$) or Earth-mass planets at the same orbital distance up to the same inclination?
 - c3) We now decide to search for Earth-twins in the habitable zone (HZ). Assume that the HZ for a G-type star (like our Sun) is approximately between 0.8 AU and 1.5 AU. What is the minimal accuracy we would have to reach to find an Earth-twin within the HZ around a sun-like star? Assume inclinations up to 45° .
 - c4) In order to make our life easier, we decided to target smaller stars now, since not only their HZ is closer to the star (between 0.25 AU and 1.3 AU for an M5-star), but the RV signal is also easier to measure. What accuracy do we have to reach to detect our Earth-twin within the HZ of an M5-star ($M = 0.21 M_\odot$)?

Problem 12.4 Transit Duration, Impact Parameter, and Rossiter–McLaughlin Effect

a) Transit Duration

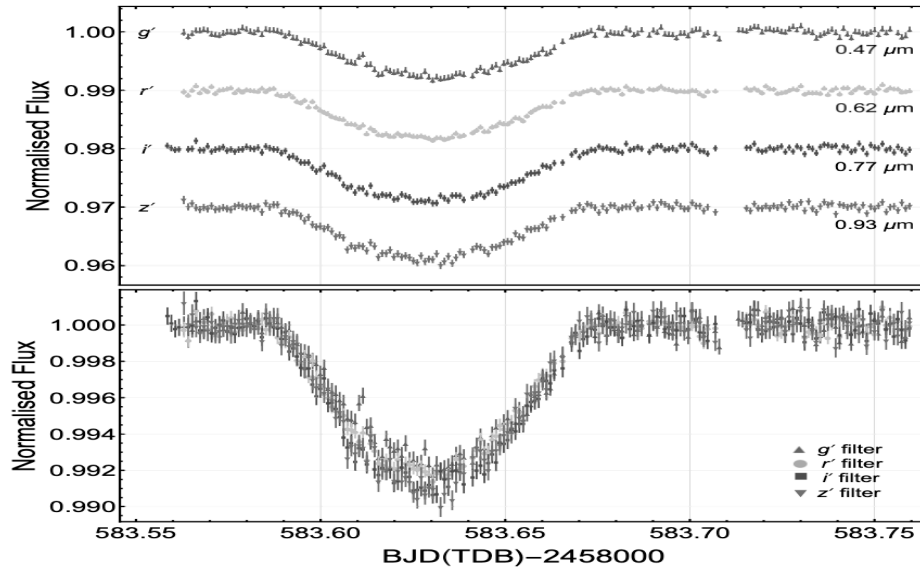
Assuming a transiting planet on a circular orbit with period P , show that the transit duration is

$$\Delta T = \frac{R_* P}{\pi a} \sqrt{1 - b^2},$$

where b is the impact parameter, R_* is the stellar radius, and a is the orbital semi-major axis.

b) Impact Parameter from Light Curves

The data points in the figure below show the normalised flux as a function of time (BJD TDB) measured at several wavelengths for a transiting planet.



Using the figure, show that the impact parameter of the transit is $b = 0.92$, given:

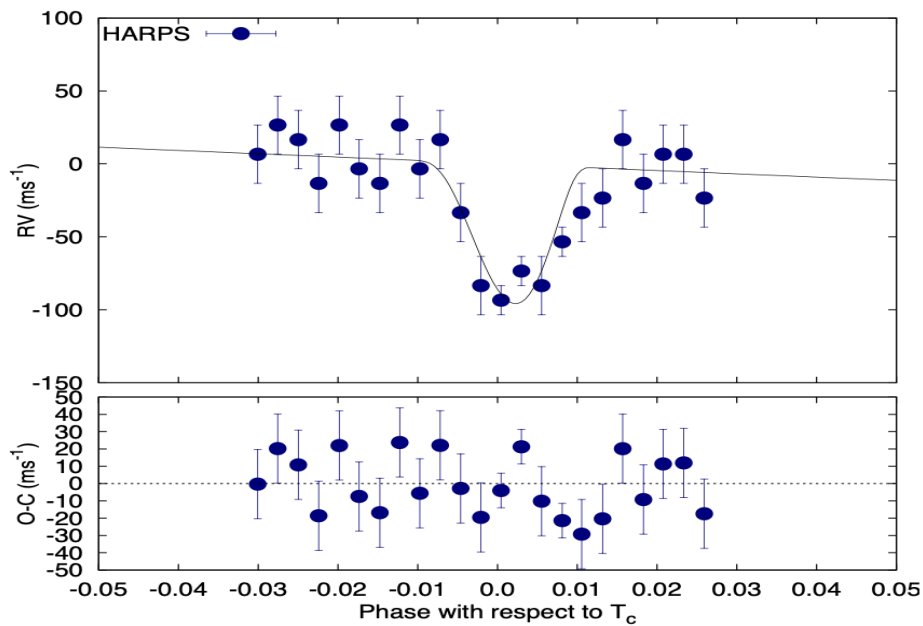
- Stellar mass: $M_* = 1.24 M_\odot$
- Stellar radius: $R_* = 1.35 R_\odot$
- Orbital period: $P = 4.23$ days

Comment on and justify which wavelength channel should be used to maximise the accuracy of the b measurement.

Hint: $R_\odot = 0.7 \times 10^9$ m.

c) Rossiter–McLaughlin Effect

The figure below shows radial-velocity measurements taken during the transit above as a function of orbital phase, with $T_c = 0$ at mid-transit.



Using these measurements together with the impact parameter derived in Part (b), sketch a physical representation of the transit superimposed on the stellar disk. Clearly identify the trajectory of the planet, the direction of motion, the orientation of the stellar rotation axis, and the direction of stellar rotation.

d) Planetary Equilibrium Temperature and Occultation Depth

Assume the planet's surface behaves as a perfect black body, so that its luminosity is $L = 4\pi\sigma R^2 T^4$.

1. Show that the equilibrium temperature of the planet satisfies $T_p^4 = T_*^4 (R_*/2a)^2$.
2. Taking the stellar surface temperature $T_* = 6400$ K, compute T_p .
3. Hence determine the occultation depth observable at $\lambda = 20\ \mu\text{m}$.

Hint: $L = 4\pi d^2 F$, where F is the flux at distance d from the source.